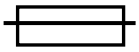
# 11 PLUS PREPARATION

## NON-VERBAL REASONING PAPER

| | |
|---|---|
| **Name:** | |
| **Date of Birth:** | |
| **Today's Date:** | |

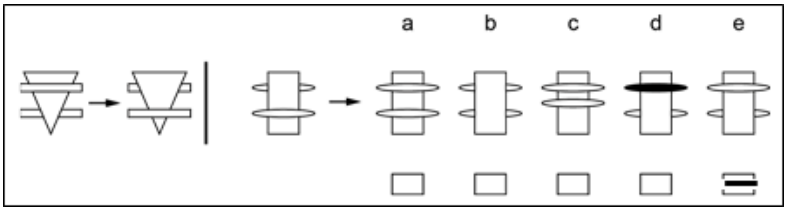
## READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 25 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 39 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

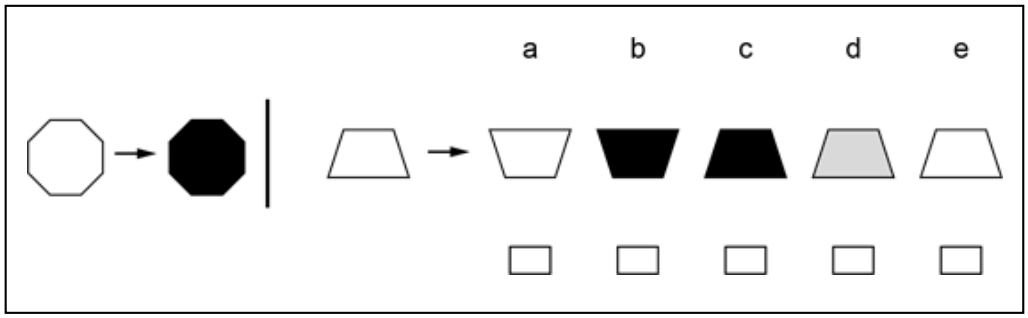
## 1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

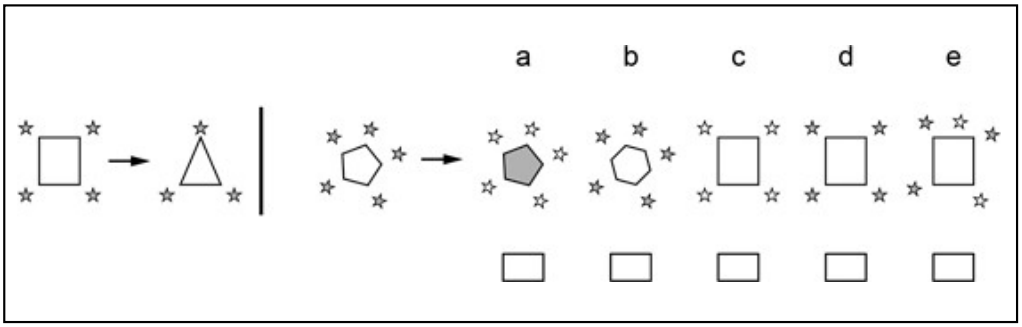
Example



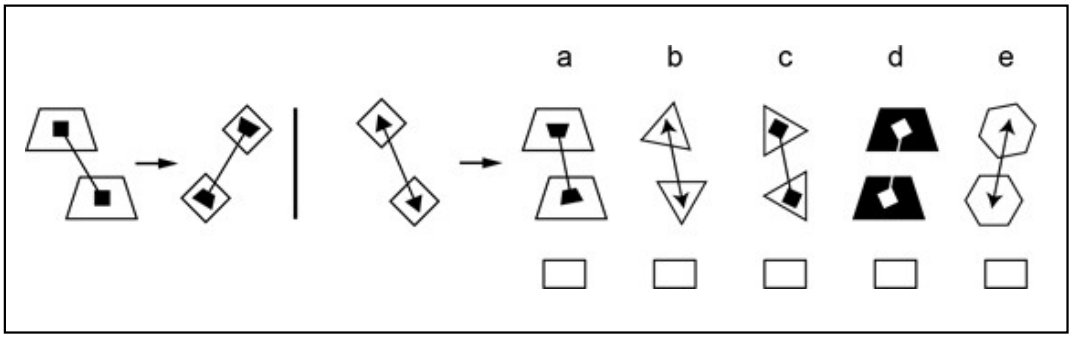

1




2



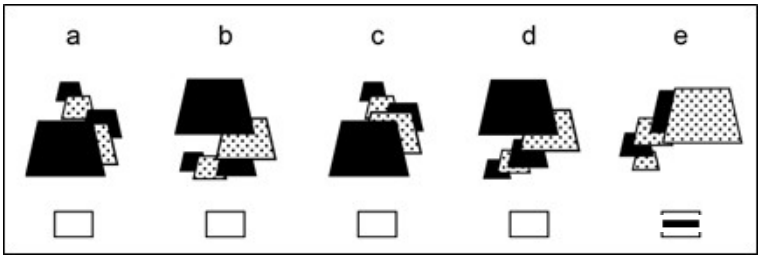

3



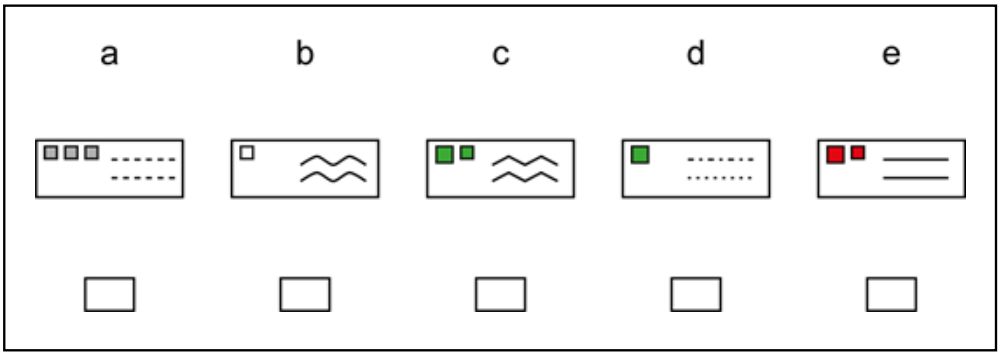
## 2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

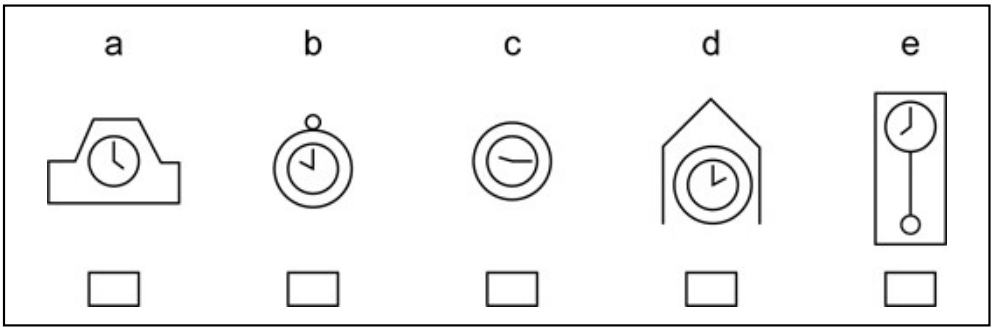
Example



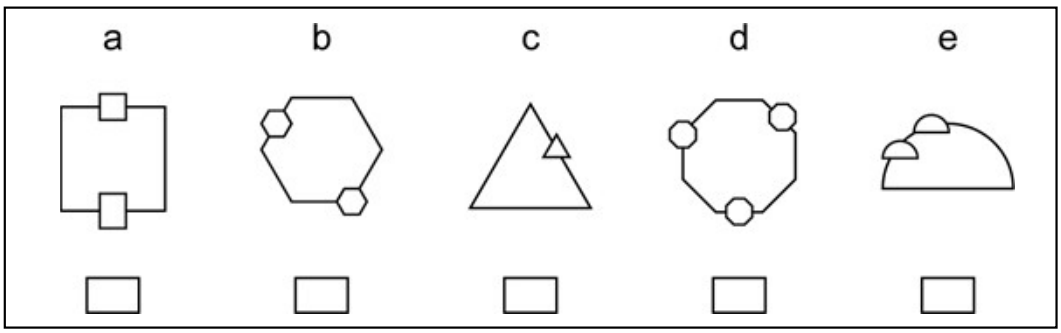

1




2



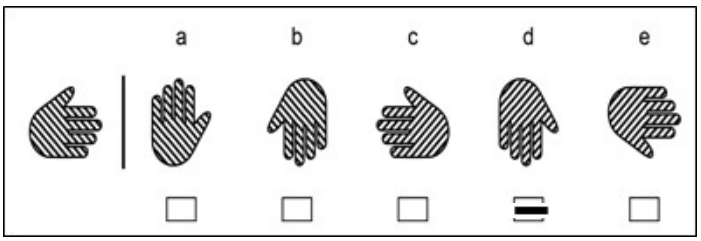

3



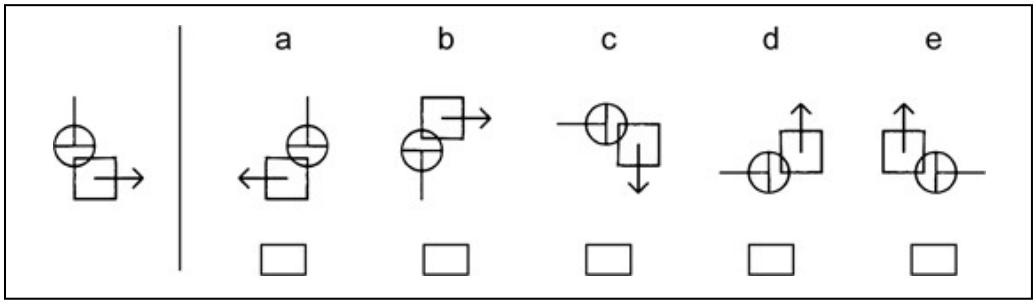
## 3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

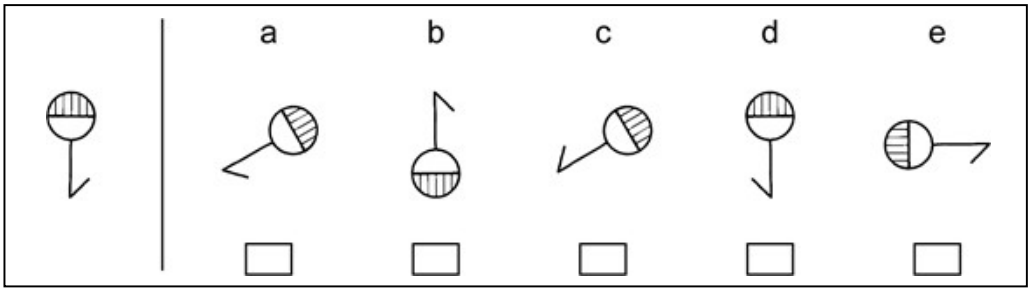
Example



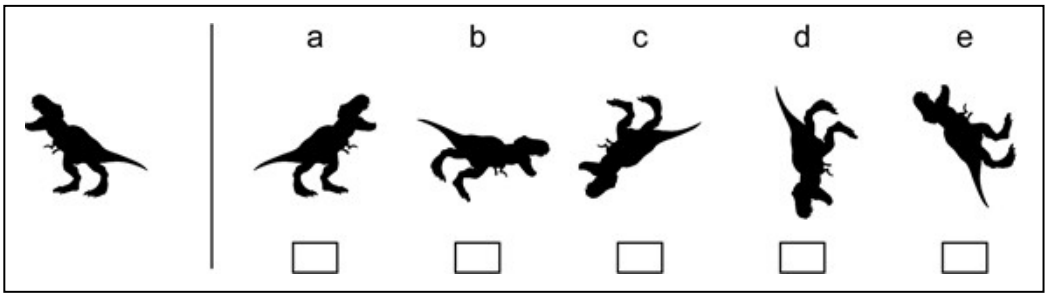

1




2



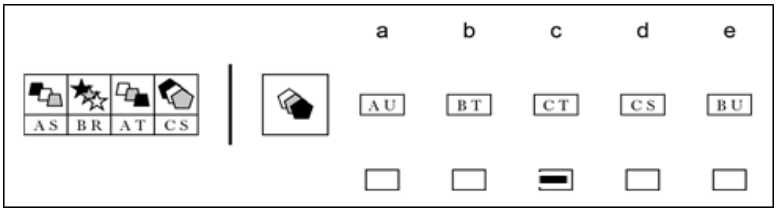

3



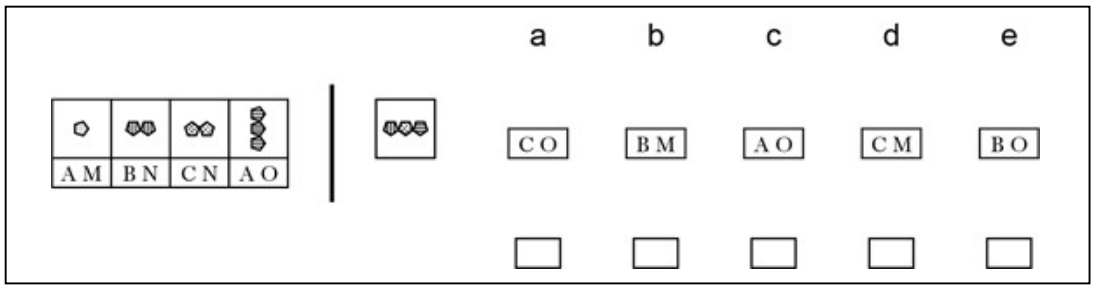
# 4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

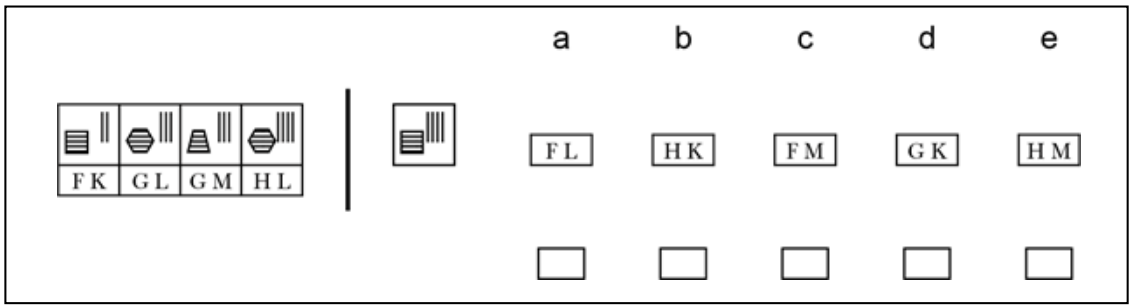
Example



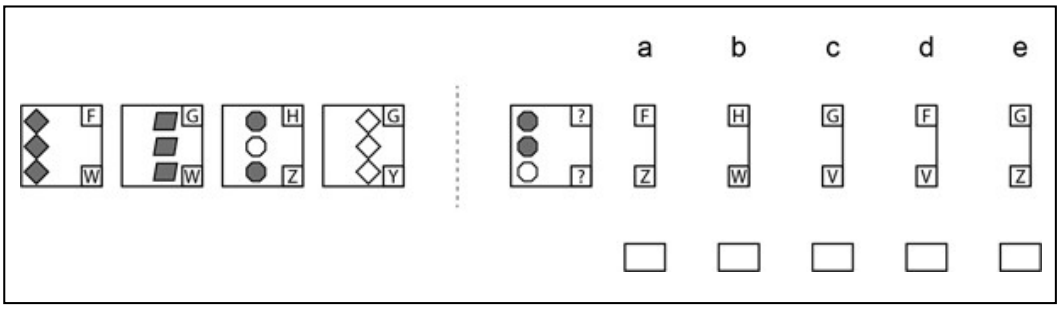

1




2



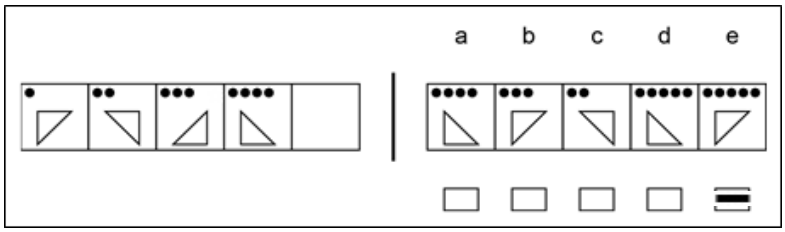

3



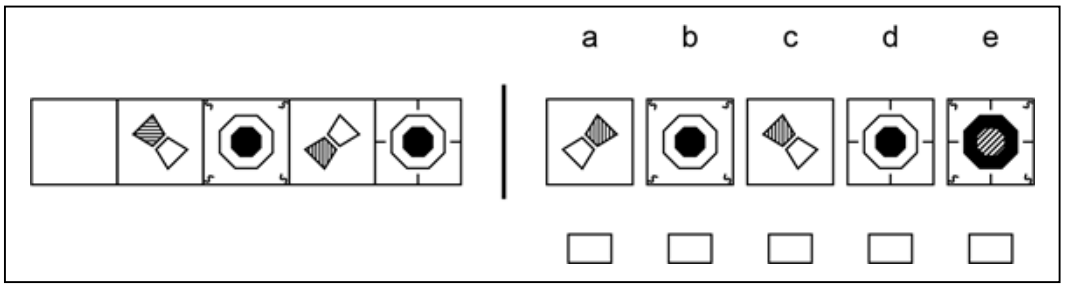
## 5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

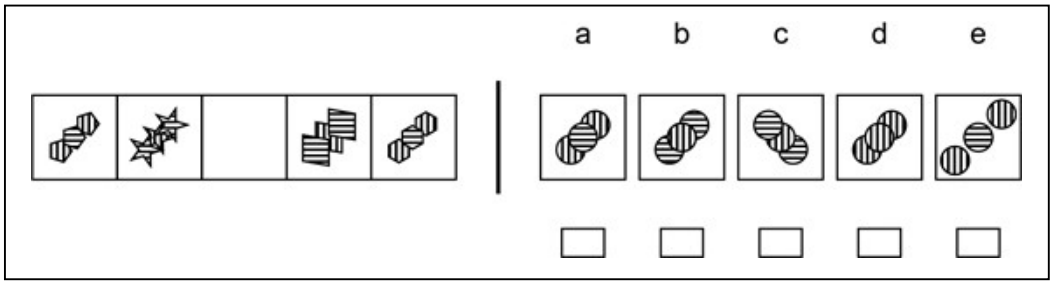
Example



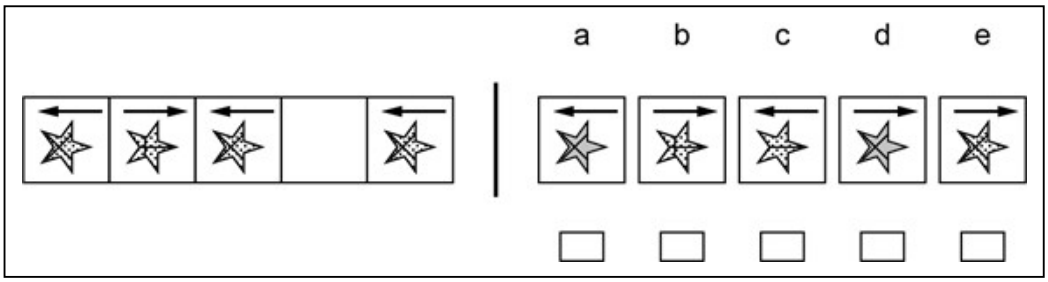

1




2



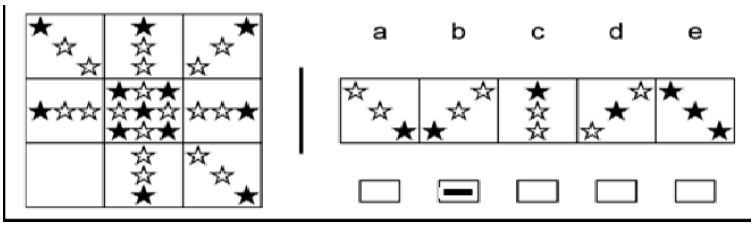

3



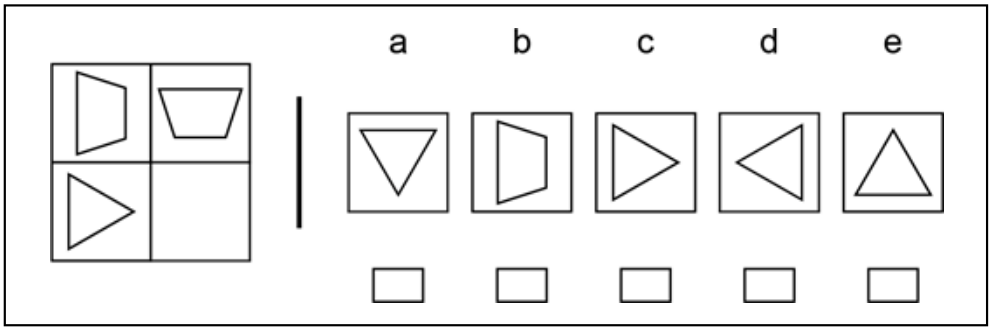
## 6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

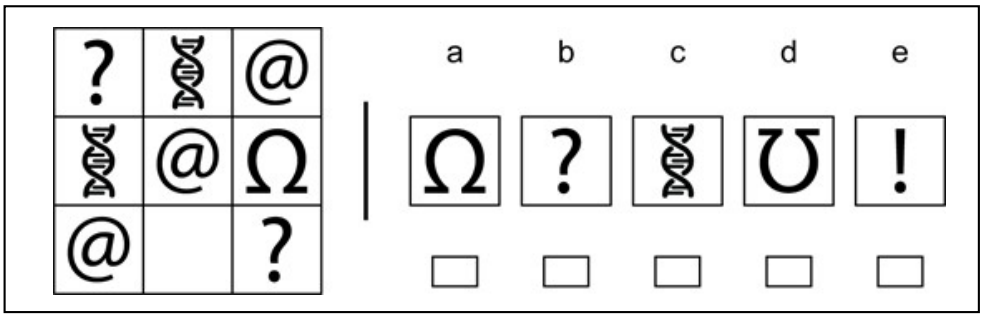
Example



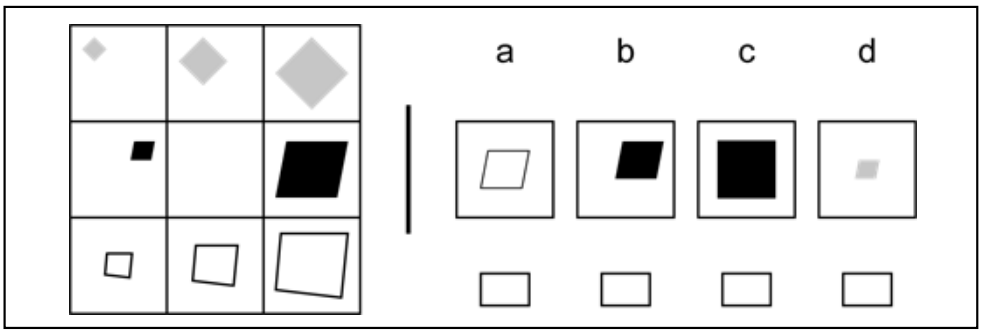

1




2



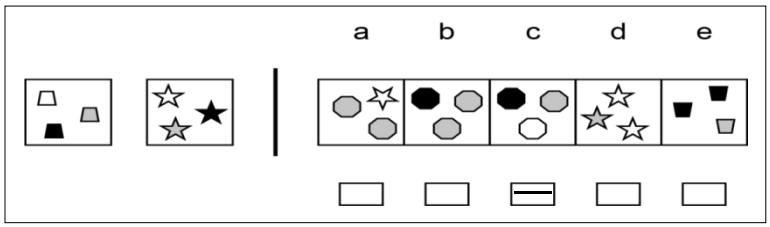

3



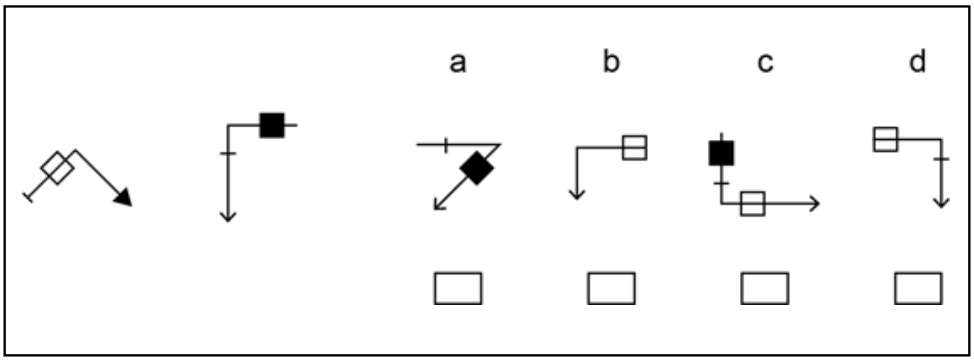
## 7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

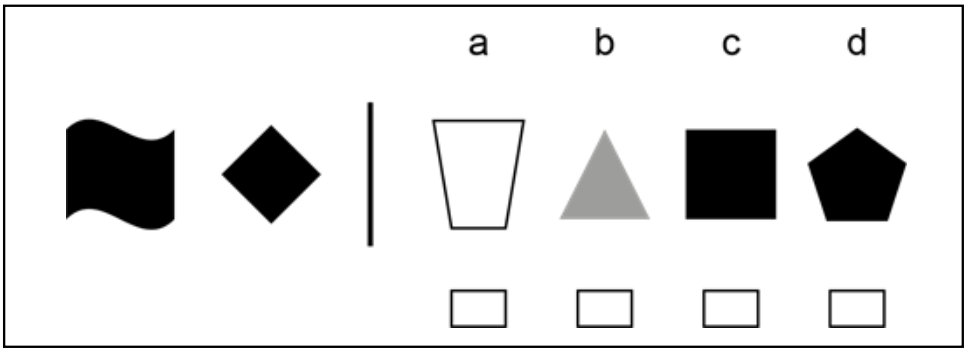
Example



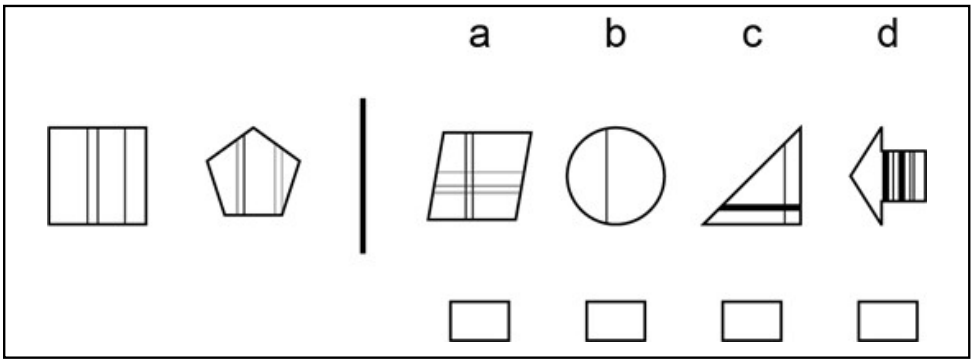

1




2



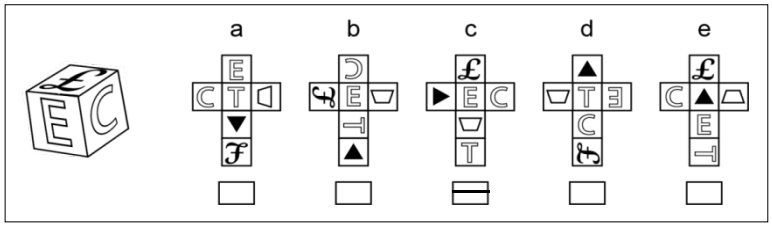

3



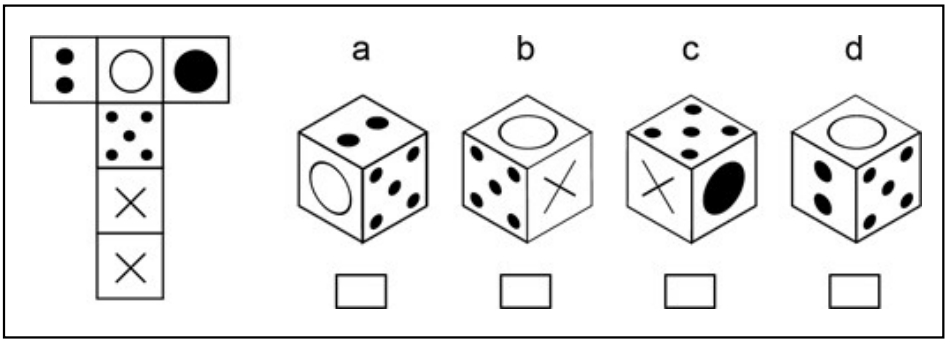
## 8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

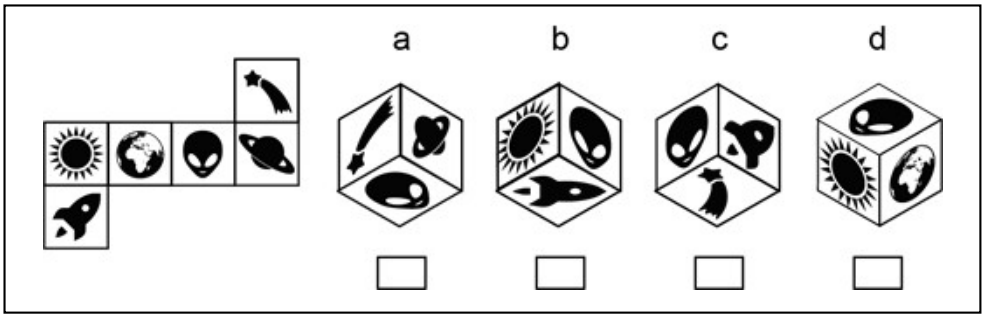
Example



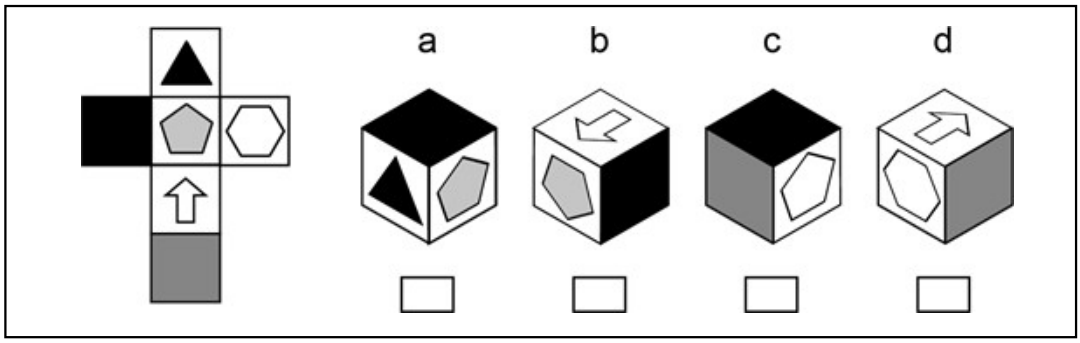

1




2



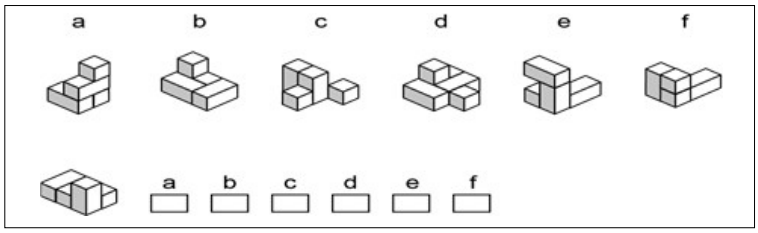

3



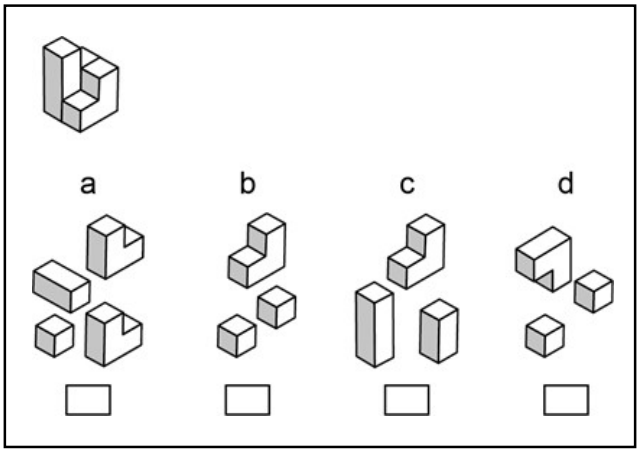
## 9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

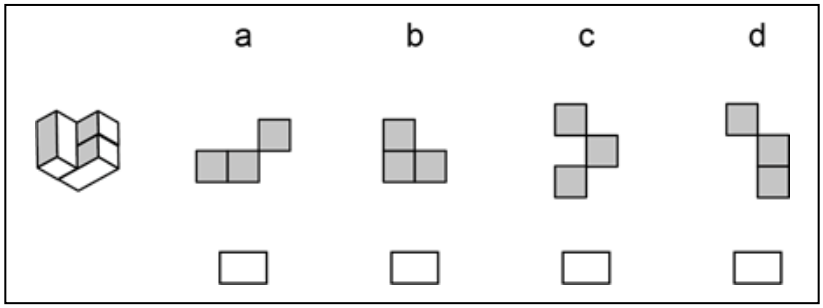
Example



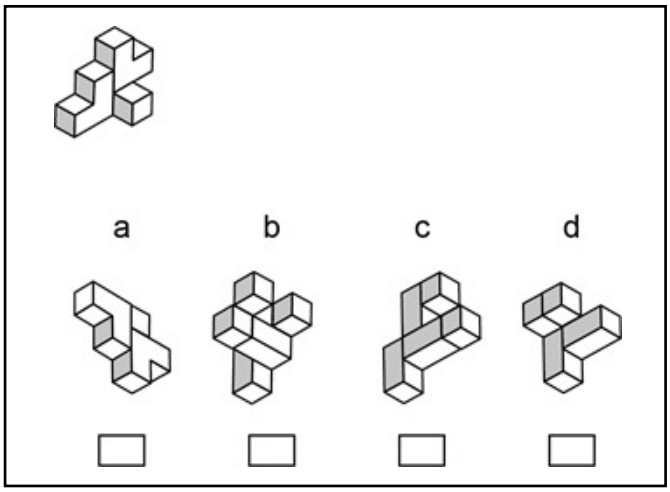

1




2



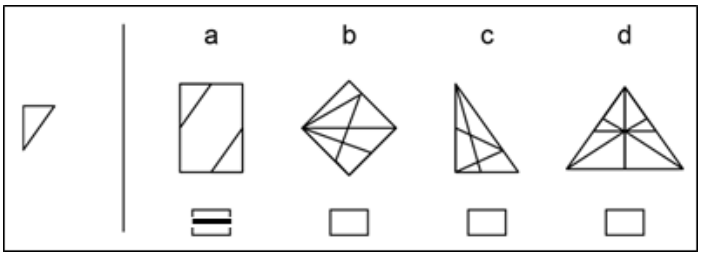

3



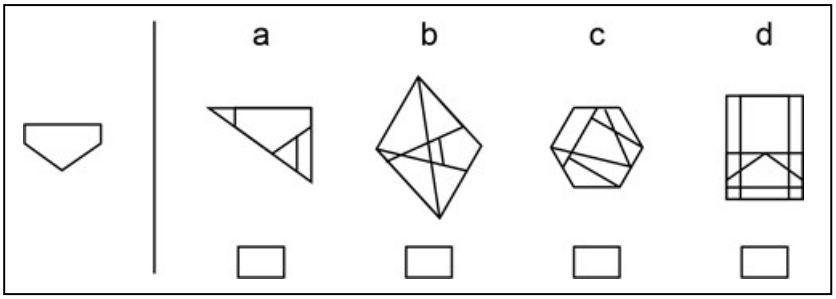
## 10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

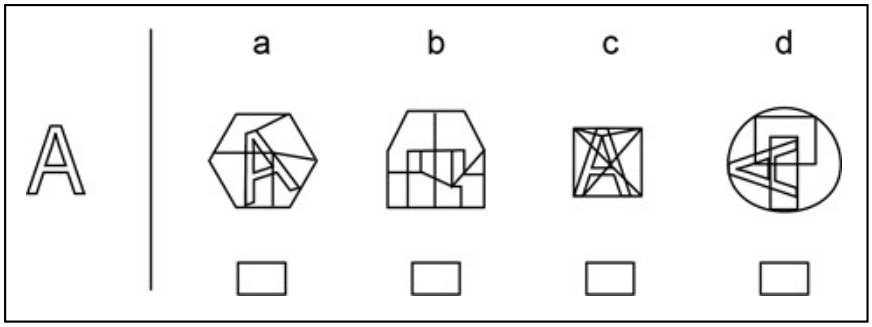
Example



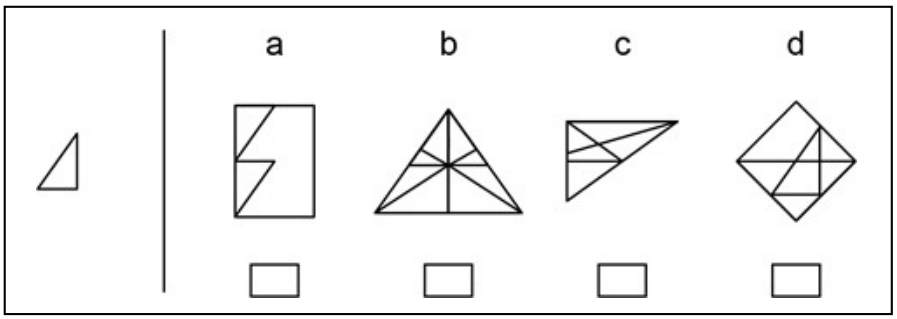

1




2



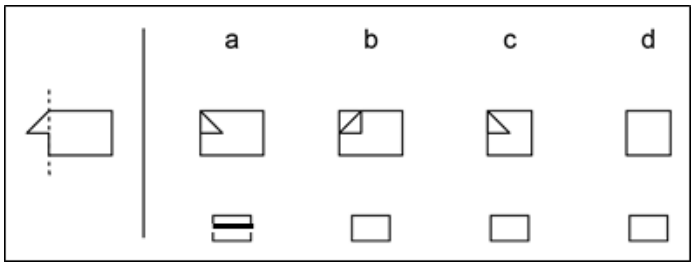

3



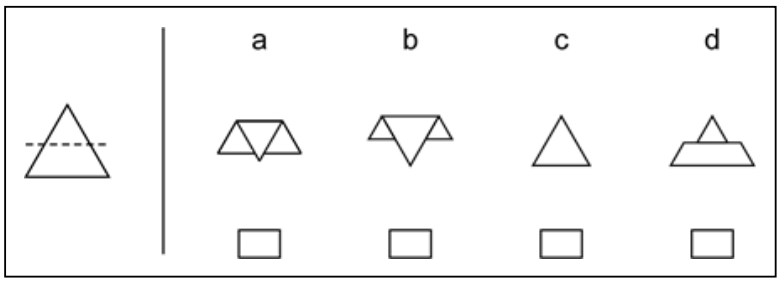
# 11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

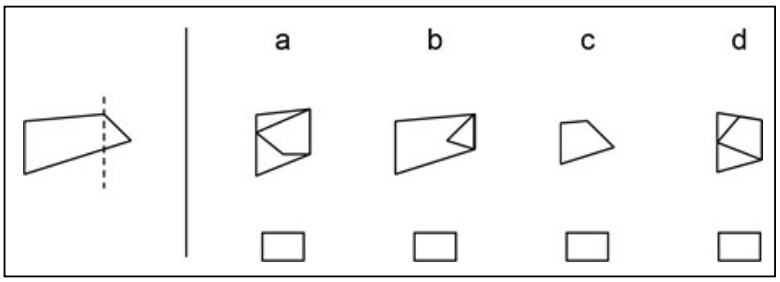
Example



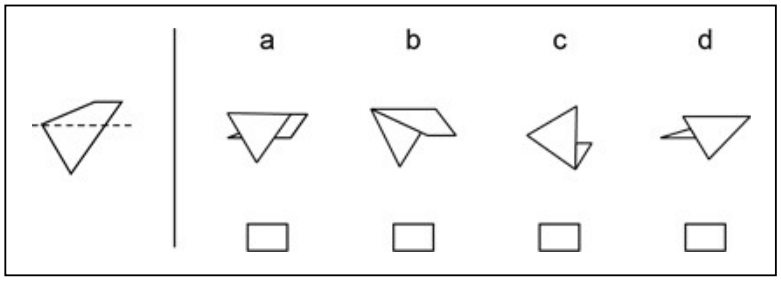

1




2



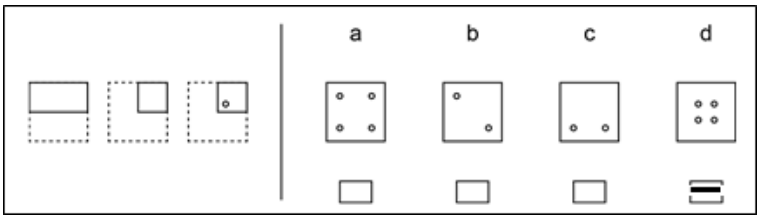

3



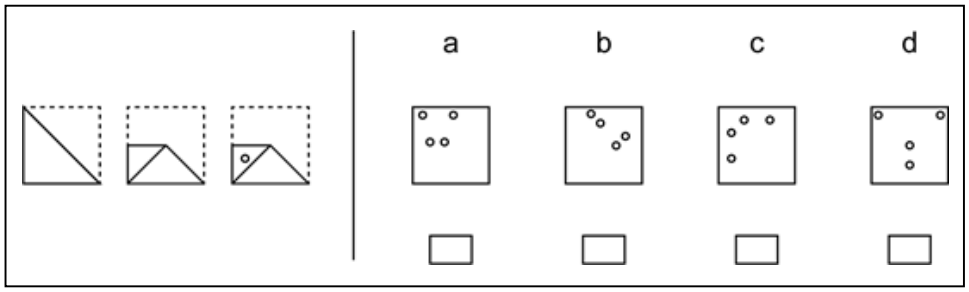
## 12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

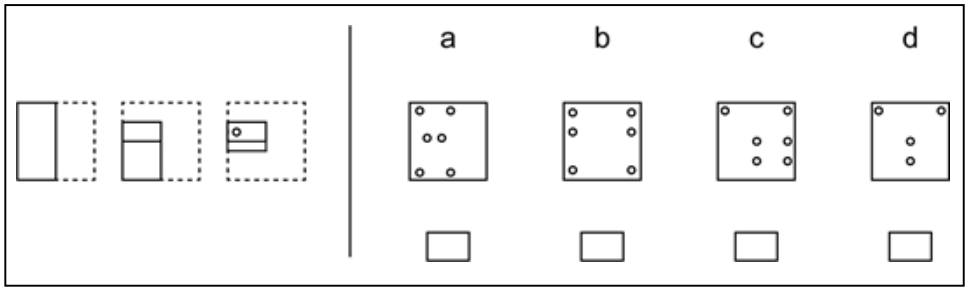
Example



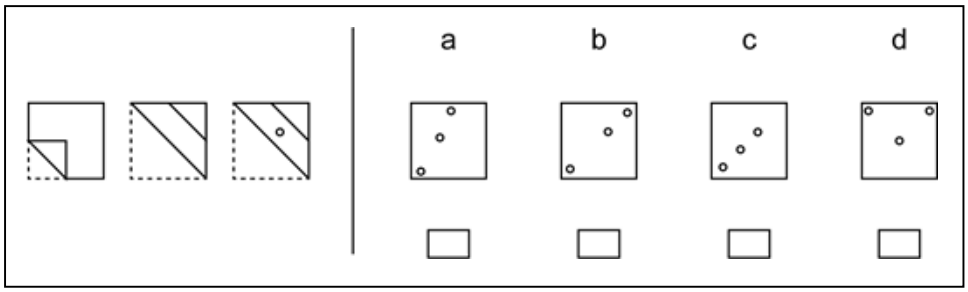

1




2



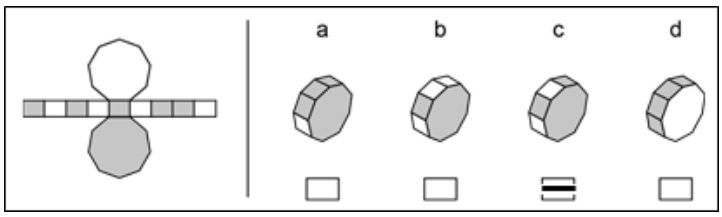

3



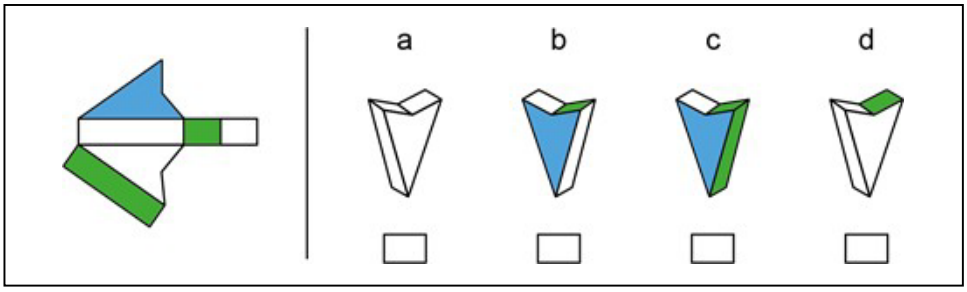
## 13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

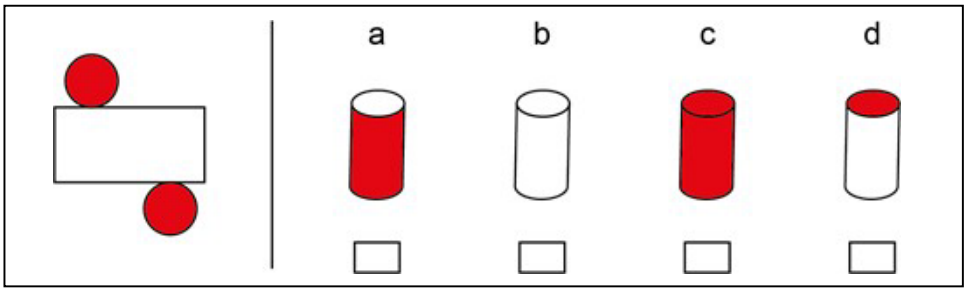
Example




1




2

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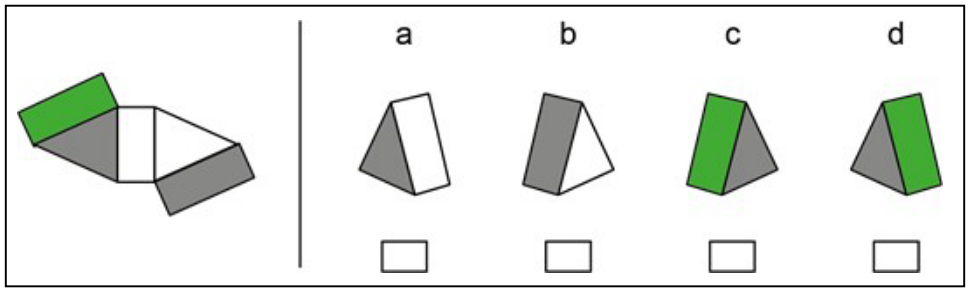


## Non-Verbal Reasoning

### Like Shapes

1. **C** - (The shape simply changes from white to black.)
2. **D** - (The first shape has the same number of sides and stars. The second shape then has one less side and so one less star so the number of sides and stars are the same. A and C are incorrect because they do not follow the same shading pattern.)
3. **C** - (The inside black shape that is attached to the line becomes the large outer white shape and the outer shape shrinks and becomes the small black shape attached to the line inside.)

### Odd One Out

1. **D** – (All answer options have a pair of identical lines to the right of the shape apart from D, which has one line of dots and dashes and one line of just dots.)
2. **C** – (All answer options show the time as being on the hour apart from C, which is showing quarter past.)
3. **A** – (All answer options show two smaller identical shapes on the perimeter of a larger shape of the same type apart from A, which has one square and one rectangle on the perimeter of a square.)

### Rotations

1. **D** – (D is a rotation of 90 degrees anti-clockwise.)
2. **A** – (A is a rotation of approximately 60 degrees clockwise.)
3. **D** – (D is a rotation of approximately 135 degrees anti-clockwise.)

### Code Breaker

1. **E** - (You can work out that A stands for the pentagons pointing to the right because there are two of them. Therefore, B stands for the pentagons pointing down and C stands for the pentagons pointing up. In the right hand letters there are two Ns and the similarity between the two is that there are two pentagons in each square. So, O stands for three pentagons in a square. Therefore the final answer is BO.)

2. **B** - (You can work out that G stands for there being three vertical lines as there are two of them. Therefore, F stands for there being two vertical lines and H stands for there being four vertical lines. In the right hand letters there are two Ls and the similarity between the two is that both of the shapes are hexagon. So, K must be the shape being a square. So, the final answer is HK.)
3. **A** - (You can work out that W stands for all three shapes being shaded as there are two of them. Therefore, Z stands for two shapes being shaded and Y stands for none of the shapes being shaded. There are two Gs in the top letters and the similarity between the two is that the shapes are on the right hand side of the square. Therefore, F stands for the shapes being on the left hand side of the square. So, the final answer is FZ.)

**Missing Sequence**

1. **D** - (The second and fourth squares provide no information to answer the question so can be ignored. Looking at the third and fifth squares, there is a large white octagon with a smaller black octagon inside, with some lines then on the edge of the square. The only answers with these requirements are B and D so it is then clear the odd number squares alternate between the straight lines on the sides and the wiggly lines in the corners.)
2. **A** - (As you move from left to right the order of the shapes alternates between vertical lines, horizontal lines, vertical lines and horizontal lines, vertical lines, horizontal lines. Also, the middle shape alternates between being in front of the other two shapes and being behind the other two shapes. In the missing square the shapes should alternate between vertical, horizontal, vertical lines and should have the middle circle in front of the other two, so the answer must be A.)
3. **B** - (The squares keep alternating with two changes each time. The arrow alternates between pointing left and pointing right and the line inside the star alternates between being a X and being a +. The missing square should have the arrow pointing right and the line should be a + so, must be B.)

## Missing Square

1. **A** - (In the top row you can see that the shape on the left has been rotated 90 degrees clockwise to form the shape on the right. If you rotate the shape on the left in the bottom row 90 degrees clockwise you will find that the answer is A.)
2. **A** - (The square forms a pattern in with the same shapes in a diagonal line.)
3. **B** - (As you go from left to right in each row the same shape gets bigger each time.)

## Most Like

1. **D** – (Both examples feature an arrow with a right angle. They also feature a square and a line on the arrow.)
2. **C** – (Both example shapes have four sides and are shaded black.)
3. **C** – (The number of lines inside the shape = the number of sides of the shape -1.)

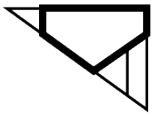
## Nets

1. **C** - (From the net you can see that A and D cannot be correct as the two black dots are in the wrong orientation with respect to the white circle. B cannot be correct as the white circle, five dots and X all appear in a vertical line in the net therefore cannot possibly fold around to meet each other in that orientation. From the net you can see that when folded down the black circle will share an edge with both the white circle and the five dots. Therefore C is the correct answer.)
2. **A** – (From the net you can see that the tail of the shooting star shares an edge with the ringed planet and that when it folds around the star part of the shooting star will share an edge with the top of the alien's head.)
3. **B** – (From the net you can see that the top of the arrow shares an edge with the base of the pentagon. When folded around the black face will be to the left of the pentagon and to the left of the arrow.)

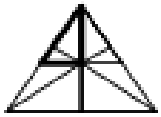
## 3D Shapes

1. **C** – (You can see from the 3D shape that it is made up from one L shaped block, one longer cuboid and one medium length cuboid of the same length as the L shaped block.)
2. **B** – (When viewed in 2D the shape could either appear as a square or L shape depending on the angle. As a square is not one of the answer options you know the answer must be B.)
3. **A** – (The shape has been rotated so that it is now upside down.)

## Hidden Shape

1. **A** – 

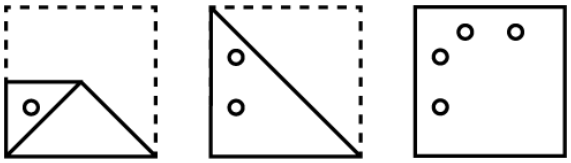
2. **C** -

3. **B** – 

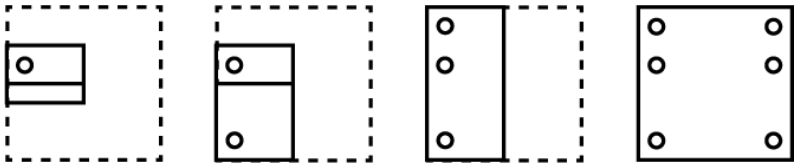
## Fold Along the Dotted Line

1. **A** – (The top portion of the shape will fold down over the bottom portion.)
2. **B** – (The right portion of the shape will fold over the left portion.)
3. **B** – (The top portion will fold over the bottom portion so that the left part of the bottom portion is still visible underneath.)

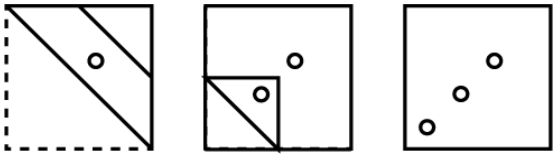
## Hole Punch

1. **C** - 

2. **B** -



3. **C** –



**Shaded Nets**

1. **B** - (From the net you can see that the rectangular faces will alternate in colour between green and white, with the small green rectangular face being at the top right if looking at the blue side. B is the only option that satisfies this.)
2. **D** - (You can see from the net that the circular faces must both be red, ruling out options A and B. C cannot be correct as the main body of the cylinder must be white.)
3. **D** – (When the net is folded to make the shape the longer rectangular face will either be green or grey, ruling out option A. The answer cannot be B or C as the grey and green rectangular faces are on the wrong side of the respective triangular faces.)